

BACT7507 – Project

Week 1 Lab Activity – Project Ideation & Planning Workshop (3 Hours)

Lab Objectives

- Identify a real-world IT problem.
- Analyse existing solutions.
- Define realistic project scope.
- Draft initial project idea.
- Set up GitHub repository for version control.

Part 1 – Individual Problem Brainstorm (30 minutes)

Identify THREE real problems. For each problem answer:

- Who experiences this problem?
- Why does it matter?
- What happens if it is not solved?
- Rules: Must be realistic. No large-scale AI or social media clones.

1) Eye Glass Store where users will be able to order glasses, use their camera to automate the best pair of specs based on their preferences and face.

❖ Who experiences this problem?

Mainly GenZ.

❖ Why does it matter?

It really helps in finding what suits your face while supporting their eyesight.

❖ What happens if it is not solved?

People can struggle with keeping up with their glasses as later it could be a bit too big/small on their face.

2) To give automated recipe with the ingredients a user has in their pantry/fridge

❖ Who experiences this problem?

- **Parents(working).**
- **Young adults.**
- **Students.**

❖ Why does it matter?

It not only saves time but makes use of the available resources around them.

❖ What happens if it is not solved?

It becomes a hustle to plan a quick meal after a tiring day or when you are in a rush.

3) To generate a simple interior design based on the options given by user (color, theme, furniture)

❖ Who experiences this problem?

New homeowners or someone who is either renovating/doesn't like their current state.

❖ Why does it matter?

It helps in giving a totally new vibe.

❖ What happens if it is not solved?

It becomes a struggle to come up with a idea of how things may turn out and hiring a interior designer is expensive.

Part 2 – Existing Solution Analysis (30 minutes)

For your BEST problem idea:

- Find 1–2 existing apps/tools solving similar problem.
- What features do they have?
- What do they do well?
- What is missing?
- How can your solution improve it?

1) **Eye Glass Store** where users will be able to order glasses, use their camera to automate the best pair of specs based on their preferences and face.

❖ Find 1–2 existing apps/tools solving similar problem.

Specsavers (website).

❖ What features do they have?

It basically just layouts the glasses they have available and you can try it on to see our it looks on you by turning on the camera.

❖ What do they do well?

Specsavers provides professional eye care, hearing care, and retail optical services, offering comprehensive eye tests, prescription glasses, contact lenses, and hearing aids.

❖ What is missing?

I feel like it should suggest what looks good on you rather than going through each spec available.

❖ How can your solution improve it?

We can add features that will not only find them the perfect specs that will look good on their face card but also suggest tips to take care of their eyesight.

Part 3 – Problem Statement & SMART Objectives (30 minutes)

Write your problem statement using this template:

"[Target users] struggle with [specific issue] because [reason], resulting in [impact]."

Define SMART objectives for your project (Specific, Measurable, Achievable, Relevant, Time-bound).

People who struggle with finding the right spec because of budget and influence of others, resulting in having something that does not look good on them."

Define SMART objectives for your project (Specific, Measurable, Achievable, Relevant, Time-bound).

Part 4 – Define Realistic Scope (30 minutes)

- List 3–5 features to INCLUDE.
 - ❖ It will suggest to you what will suit your face type.
 - ❖ It will find specs that are within the person's budget.
 - ❖ It will track your eye appointments.
 - ❖ It will also suggest methods to keep your eyes healthy.
 - ❖ It can store your eye test results.
- List features you will NOT include.
 - ❖ Payment system.
- Ensure scope is achievable within one semester by two students.

It will be manageable

Part 5 – Feasibility & Risks (15 minutes)

- Is the required technology within your skill level?

Yes

- Can it be completed in 12 weeks?

Yes

- Identify at least TWO risks and possible mitigation strategies.
 - Handing facial measures
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Part 6 – GitHub Setup (30 minutes)

- Create a GitHub repository named: BACT7507_Project_YourName
- Add README.md and your initial idea document.
- Make your first commit and push to GitHub.
- Take screenshot of repository and commit history.

Final Submission (Upload to Canvas)

- Problem ideas document.
- Existing solution analysis.
- Problem statement & SMART objectives.
- Scope definition.
- Feasibility & risks.
- GitHub setup screenshot.